Question 1:

(a)

$$-\frac{V_{in}}{RC} = -5.6\frac{V}{ms} \Rightarrow V_{in} = 5.6 \times 2.5 \times 10^3 \times 2 \times 10^{-6} \times \frac{1}{10^{-3}} = 28\,V$$

$$\frac{V_{ref}}{RC} = 8\frac{V}{ms} \Rightarrow V_{in} = 8 \times 2.5 \times 10^3 \times 2 \times 10^{-6} \times \frac{1}{10^{-3}} = 40\,V$$

(b)

$$t = t_1 + t_2 = t_1(1 + \frac{V_{in}}{V_{ref}})$$

Now, $V_1 = -56 = -\frac{V_{in}}{RC}t_1 \Rightarrow t_1 = 10\,ms$

$$\therefore t = 17\,ms$$

$N = \frac{V_{in}}{V_{ref}}2^4 - 1 = 10.2$. So, the binary output is $(1010)_2$.

(c)

$$V_{out} = \frac{R_f}{2R}V_{ref}(1 + 0 + 0.25 + 0) = 25\,V$$

(d)

$$\Delta = \frac{R_f}{2R}\frac{V_{ref}}{2^3} = 2.5\,V$$

$$V_{out_{max}} = \frac{R_f}{2R}V_{ref}(1 + 0.5 + 0.25 + 0.125) = 37.5\,V$$

(e) Error $= \frac{V_{in} - V_{out}}{V_{in}} \times 100 = 10.71\%$

Question 2:

(a)

$$\Delta_{ADC} = \frac{16 - 0}{2^3} = 2\,V$$

$$\Delta_{DAC} = \frac{R_f}{R}\frac{V_{ref}}{2^2} = 1\,V$$

(b) (i) 4 V — 6 V
(ii) Lowest output from the DAC = 0 V. Step size = 1 V. Thus, outputs within the range in (i) are 4, 5 & 6 V. They are the outputs of '100', '101' & '110' respectively.

(c)

$$R_f' = 2R_f = 20\,k\Omega$$

$$\Delta'_{DAC} = \frac{20}{10}\frac{4}{4} = 2\,V$$

(d) $n_R = 2^2 = 4$, $n_{OA} = 2^2 - 1 = 3$. Encoder: 4 to 2.

Question 3:

(a) $V_{TH} = \dfrac{2}{2+3}15 = 6\,V, \quad V_{TL} = \dfrac{2}{2+3}(-15) = -6\,V$

$$T_1 = R_x C_x \ln\left(\frac{-6-15}{6-15}\right) = R_x C_x \ln(6) \Rightarrow \frac{1}{2\times 500} = 22\times 10^3 \ln(6)\times C_x \Rightarrow C_x = 53.65\,nF$$

$$\tau = R_x C_x = 1.18\,ms$$

(b)

$$T_1 = R_x C_x \ln\left(\frac{V_{TL}-V_H}{V_{TH}-V_H}\right) = R_x C_x \ln\left(\frac{\frac{R_1}{R_1+R_2}V_L - V_H}{\frac{R_1}{R_1+R_2}V_H - V_H}\right)$$

$$= R_x C_x \ln\left(\frac{\frac{R_1}{R_1+R_2}(-V_H) - V_H}{\left(\frac{R_1}{R_1+R_2}-1\right)V_H}\right) = R_x C_x \ln\left(\frac{-\left(\frac{R_1}{R_1+R_2}+1\right)}{\frac{R_1}{R_1+R_2}-1}\right) = T_2$$

$$T = 2T_1 = 2R_x C_x \ln\left(\frac{2R_1+R_2}{R_2}\right)$$

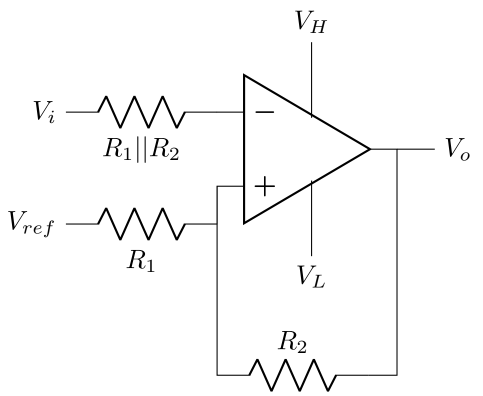
$$\Rightarrow T_{new} = 2R_x C_x \ln\left(2\frac{R_1}{R_2}+1\right)$$

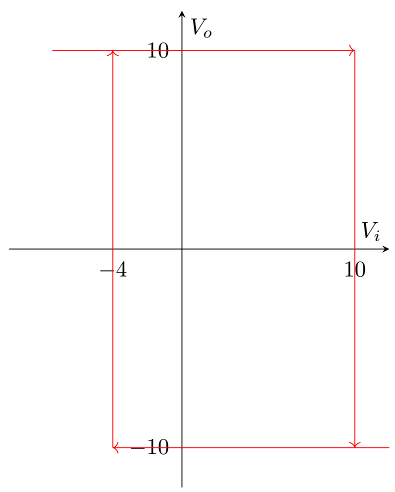
$$\Rightarrow \frac{1}{f_{new}} = \frac{1}{2f} = \frac{1}{2\times 500} = 2\times 1.18\times 10^{-3} \ln\left(2\frac{R_1}{R_2}+1\right)$$

$$\Rightarrow \frac{R_1}{R_2} = 0.26$$

(c) Inverting. $V_H = 10\,V$, $V_L = -10\,V$, $V_{TH} = 7\,V$, $V_{TL} = -7\,V$, $V_S = 0\,V$, $V_{HW} = 14\,V$

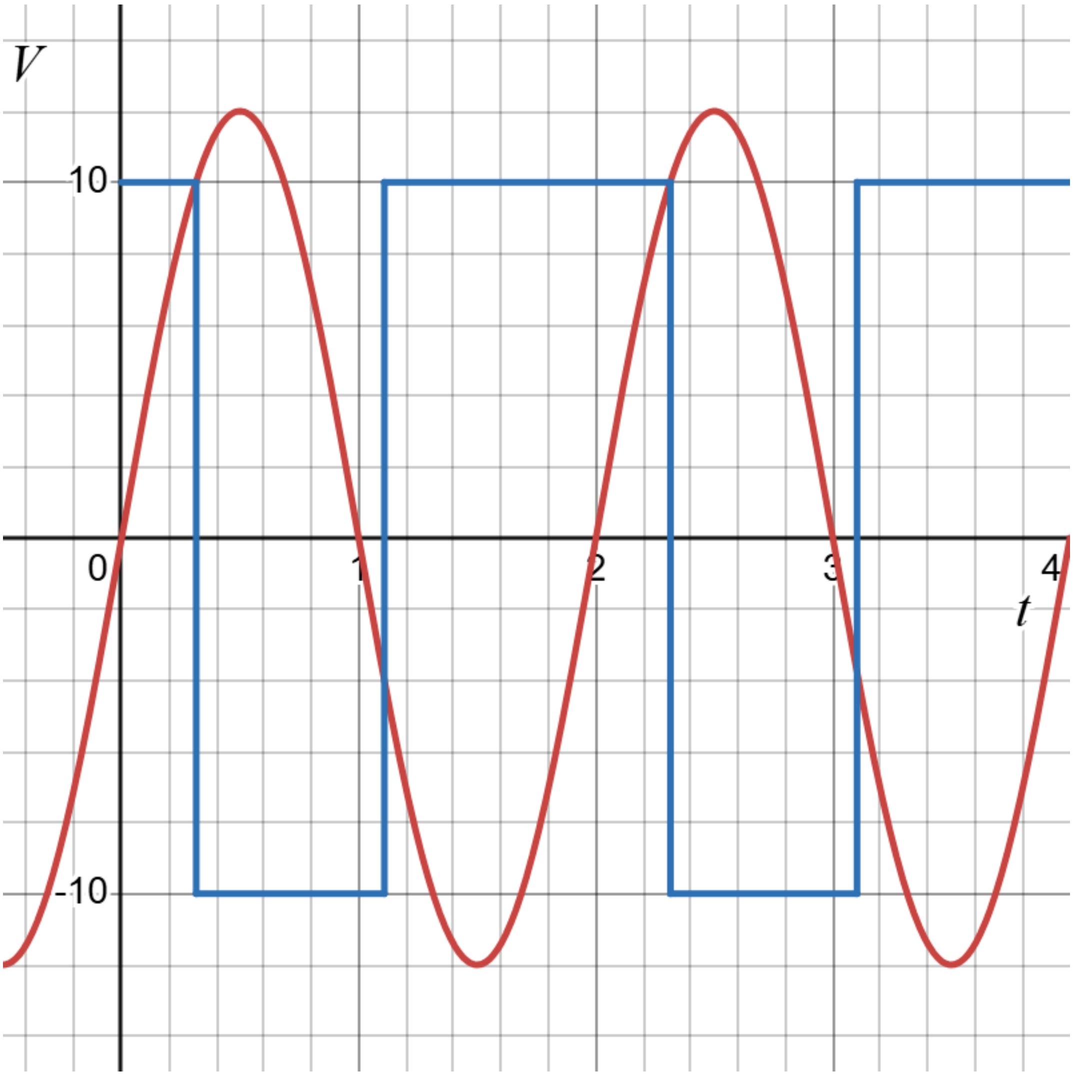
(d)





$$R_1 = 7\,k\Omega,\ R_2 = 3\,k\Omega, V_{ref} = \frac{3}{\frac{3}{7+3}} = 10\,V,\ V_H = 10\,V,\ V_L = -10\,V$$

(e)



Question 4:

(a) $T = 6\,ms$

$$V_{TH} = 4 = -\frac{R_1}{R_2}(-8) \Rightarrow \frac{R_1}{R_2} = \frac{1}{2}$$

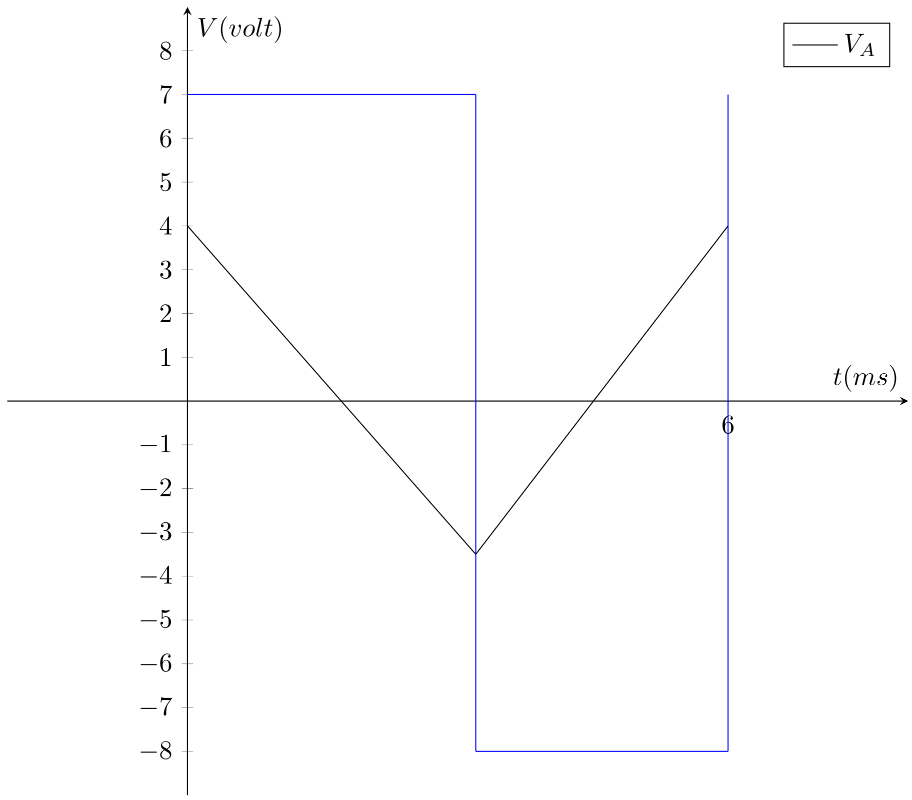
$$V_{TL} = -\frac{1}{2}7 = -3.5\,V$$

$$\therefore D = \frac{T_2}{T_1 + T_2} = \frac{\frac{V_{TL} - V_{TH}}{V_L}}{\frac{V_{TH} - V_{TL}}{V_H} + \frac{V_{TL} - V_{TH}}{V_L}} = 46.67\%$$

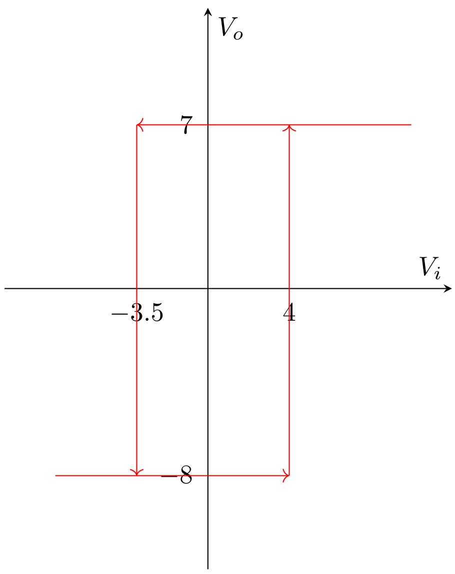
$$T_2 = D \times 6 \times 10^{-3} = 2.8\,ms$$

$$C_i = \frac{T_2}{R_i \frac{V_{TL} - V_{TH}}{V_L}} = 1.49\,\mu F$$

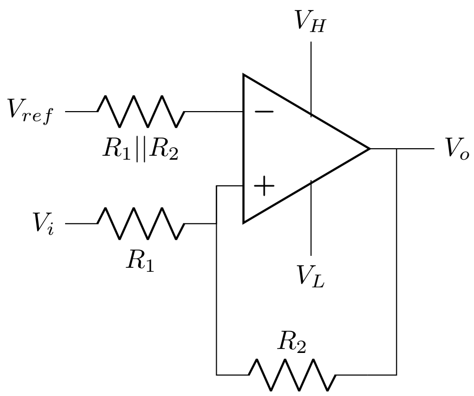
(b) $D_{sqr} = (1 - D_{tri}) \times 100 = 53.33\%$



(c)



(d)



$$V_H = 7\,V,\ V_L = -8\,V,\ R_1 = 1\,k\Omega,\ R_2 = 2\,k\Omega,\ V_{ref} = \frac{2}{\frac{1+2}{2}} = 1.33\,V$$